

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
3	(a)	(i)	Sirius A: $\lambda_{\max} = 290 \times 10^{-9}$ [m] and Canopus: $\lambda_{\max} = 400 \times 10^{-9}$ [m] [± 10 nm] for both (1) Attempt at applying $\lambda_{\max} T = 0.0029$ to both stars, even if powers of 10 incorrect (1) No need to change unit of λ to m Correct application, either by confirming Wien constant or star temperatures or $\lambda_{\max}$ (1) Accept correct calculations of $\lambda_{\max}$ even if no reference is made to the graph Application of Wien's law to one star only award 1 mark only			3	3	2	
		(ii)	Sirius A (1) Greater spectral intensity (at 'blue' end or at shorter wavelengths or towards 400 nm) (1) Don't accept peak wavelength of Sirius A is closest to the blue end of the spectrum than Canopus. Don't accept reference to temperature by itself e.g. Sirius has a higher temperature so therefore must be bluer. Accept ratio of $\frac{B}{R}$ for both stars, e.g. $\frac{6.1}{1.7}$ against $\frac{1.6}{0.8}$		1	1	2		